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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B. Tech (CBCS) Degree Examinations

December 2025

BT2205-1 – NUTRACEUTICALS

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The first FOSHU (Foods for Specified Health Uses) product approved in 1993 was:
A) Fortified bread
B) Hypoallergenic rice
C) Omega-3 enriched milk
D) Probiotic yogurt
2. Selenium acts as a nutraceutical because it is a:
A) Prebiotic fiber
B) Synbiotic compound
C) Component of glutathione peroxidase with antioxidant role
D) Structural lipid
3. Which of the following is a short-chain fatty acid produced by gut microbiota?
A) Linoleic acid
B) Oleic acid
C) Stearic acid
D) Butyric acid
4. Which cereal is a rich source of beta-glucan and helps lower cholesterol?
A) Oats
B) Rice
C) Sorghum
D) Maize
5. Anthocyanins are colored pigments found mainly in:
A) White tubers
B) Red or purple fruits
C) Green leaves
D) None of these
6. Conjugated Linoleic Acid (CLA), present in milk fat, is primarily known for its:
A) Anti-inflammatory and anti-cancer properties
B) Role in increasing lactose digestibility
C) Function as a calcium-binding protein
D) Enzymatic role in fat digestion
7. Which cereal is gluten-free and recommended for individuals with celiac disease?
A) Rye
B) Wheat
C) Barley
D) Sorghum
8. The adhesion capability of a probiotic strain is important because it:
A) Increases digestibility
B) Enables colonization of intestinal mucosa
C) Enhances bile secretion
D) Reduces immunity
9. Acid tolerance of Lactobacillus is tested using:
A) Stimulated gastric juice (pH2-4)
B) Bile salt solution
C) Distilled water
D) Alcohol solution
10. Non-polar solvents are used for extracting:
A) Sugars
B) Proteins
C) Lipids
D) Amino acids
11. In purification, Rf value is defined as:
A) Time of retention/Solvent used
B) Absorbance/Wavelength
C) pH/temperature
D) Distance traveled by solute/distance traveled by solvent front

12. An example of alkaloid is:
 A) Caffeine B) Ascorbic acid
 C) Tocopherol D) Retinol
13. A key function of glucosamine and hyaluronic acid combination is:
 A) Increasing cholesterol B) Synergistic improvement of joint and skin health
 C) Blocking calcium D) Inhibiting collagen synthesis
14. Chitin is a polymer made up of repeating units of :
 A) Glucose B) Amino acids
 C) N-acetylglucosamine D) Fatty acids
15. How do plant sterols reduce cardiovascular risk?
 A) By reducing platelet aggregation B) By blocking cholesterol absorption in the intestines
 C) By increasing HDL cholesterol D) By acting as antioxidants
16. Which nutraceutical is specifically noted for lowering LDL cholesterol?
 A) Beta-glucan B) Vitamin C
 C) Iron D) Magnesium
17. Which nutraceuticals has shown potential for cancer prevention due to its antioxidant properties?
 A) Caffeine B) Iron
 C) Lycopene D) Calcium
18. Which nutraceutical has shown promise in improving insulin sensitivity?
 A) Iron B) Copper
 C) Vitamin A D) Chromium
19. Green tea extracts aids obesity control by:
 A) Increasing fat oxidation B) Decreasing metabolism
 C) Raising blood glucose D) Reducing appetite only
20. Zinc is essential for immune health because it:
 A) Lowers white blood cell count B) Raises blood sugar
 C) Supports T-cell function D) Inhibits neutrophils

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write short notes on the classification of nutraceuticals based on chemical nature.	8	L*1	1	1
	b) What are carotenoids? Describe the types, sources and health benefits of carotenoids.	8	L2	2	3,6
2.	a) Describe the sources and types of functional oligosaccharides. Elaborate on the physiological benefits.	8	L2	1	3,6
	b) Discuss the use of cereals as fermentable substrates for the growth of probiotic microorganisms and their influence on gut health and immunity.	8	L3	2	3,6
3.	a) Comment on – i. Antihypertensive peptides ii. Eicosapentaenoic acid	8	L2	1	3,6
	b) Discuss the health benefits of calcium and magnesium. Describe their common dietary sources.	8	L2	2	3,6
Unit – II					
4.	a) Describe the Stass-Otto method used for the alkaloid extraction. Add a note on its applications	8	L1	3	1,3,6

b)	Discuss common bacterial strains used as probiotics and their specific health benefits.	8	L2	4	3,6
5. a)	Explain the extraction method for chondroitin sulfate from animal sources.	8	L2	3	4,6
b)	Describe the isolation process of probiotic microorganisms.	8	L2	4	3,6
6. a)	Comment on the sources, extraction process and health benefits of Hyaluronic acid.	8	L2	3	3,4,6
b)	Describe any two methods involved in the extraction of antioxidants from algal sources.	8	L2	4	3,4,6
Unit – III					
7. a)	Describe the role of nutraceuticals in glycemic control.	8	L3	5	3,6
b)	List out eight bioactive compounds with antioxidant property which helps in cancer prevention. Add a note on their sources.	8	L3	5	3,6
8. a)	Comment on how nutraceuticals play a role in reducing cholesterol levels.	8	L3	5	3,6
b)	Analyze the impact of dietary fiber on atherosclerosis prevention and cardiovascular function.	8	L3	5	3,6

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
